

Highlights

- Inverse-propensity graph propagation is biased beyond one hop.
- The repeated-walk bias grows as the inverse of the propensity clip.
- DRUP removes this bias exactly for any number of hops, with no training.
- One structure yields fairness, explanation, robustness and privacy results.
- DRUP beats tuned GNN baselines on Yahoo! R3 and KuaiRec unbiased tests.